Partner-Specific Affective Precision in Social Active Inference

Anonymous Author(s)

Anonymous Institution

Abstract. Affective active inference formalizes valence as inference over expected policy precision: an internal estimate of how well an agent's generative model supports action. In social interaction, however, model fitness is not only global. An agent may predict one partner reliably, remain uncertain about another, and be misled by a third. We extend this formulation by factorizing affective precision across partners in a repeated trust-game simulation. The agent maintains separate predictive models for each partner; after each interaction, partner-specific prediction error updates a local precision estimate, which modulates confidence in future policies involving that partner.

We evaluate stable interaction, partner choice, and abrupt betrayal regimes. Partner-local precision changes how social beliefs are acted upon, shaping which partners the agent returns to, how decisively it acts, and how it reallocates after surprising outcomes. These effects are strongest in decision-ambiguous regimes, where multiple actions remain viable. They can also emerge even when agents with and without precision modulation maintain similar beliefs about their partners, indicating a deployment rather than knowledge difference. Because this affective signal tracks predictive reliability more than partner reward, its behavioral influence supports a model-fitness interpretation. Under abrupt betrayal, however, surprise-driven precision updates can over-sharpen decisions, locking the agent into outdated assumptions and reducing payoffs after the behavioral switch. These results suggest that extending affective inference to multi-agent systems requires localizing precision over social models, providing a computational account of how implicit metacognitive estimates of model fitness can both support and misdirect social action.

Keywords: Active inference · Affect · Trust games

1 Introduction

Active inference treats perception and action as coupled inference under a generative model [3, 4, 9]. Agents infer hidden causes of observations, evaluate policies by expected free energy, and select actions through a precision-weighted posterior over policies [5, 2]. Policy precision is therefore not just an implementation temperature. It controls whether an agent acts decisively on the currently preferred policy or preserves a diffuse distribution over several plausible actions.

Affective active-inference accounts give this precision psychological content. Hesp et al. [7] formalize valence as inference over expected action precision, or subjective model fitness: the agent estimates whether its current model is reliable enough to support confident action. This makes affect a metacognitive signal about the agent's own model rather than an externally added reward bonus.

Social interaction makes the model-fitness problem local. A focal agent may predict one partner reliably, remain uncertain about another, and be confidently adversarial toward a third. A single social precision state collapses distinctions that matter for action. A predictable exploiter can be low value but high model fitness, because the agent knows what to expect and can avoid or defect against them. A volatile ally can be high value but low model fitness, because the agent cannot yet rely on its partner model to choose safely.

We test this idea in a repeated multi-partner trust-game simulation. The agent maintains a separate discrete generative model for each partner. After an interaction, partner-action prediction error updates a partner-local beta posterior, and the expected beta modulates policy precision for future decisions involving that partner. The central claim is about *deployment*: the step that turns social beliefs into action. An agent may already maintain useful beliefs about a partner yet act on them too weakly, too diffusely, or too confidently at the wrong time.

This framing yields falsifiable alternatives. If beta is cached value, precision should track payoff more strongly than predictive reliability. If beta simply improves belief inference, lesioning the beta-to-policy pathway should damage beliefs as much as behavior. If affect is a generic performance boost, it should improve payoff across regimes. The simulations below support a narrower mechanism. Partner-local precision tracks model fitness, changes policy deployment in open regimes, and can misdirect action when abrupt volatility makes the partner model stale.

The contribution is a partner-specific extension of affective precision for social active inference. It is not a claim that affect always improves payoff, nor that beta is a literal trust scalar. The experiments show that partner-local precision changes how social beliefs are acted upon, including partner selection and post-betrayal reallocation, while also revealing a boundary condition: abrupt betrayal can make surprise-driven precision updates over-sharpen decisions around outdated assumptions.

2 Model and Experiments

We evaluate partner-specific affective precision in a repeated trust game derived from the economic trust-game tradition [1, 8]. A focal active-inference agent interacts with four partners. Each partner has a latent type (cooperator, reciprocator, exploiter, or random) and a latent stance toward the agent (trusting, neutral, or hostile). The agent observes the partner's action and its own payoff,

Table 1: Binary trust-game payoff to the focal agent. The graded task uses the same reliance structure with discretized investment levels.

	Partner cooperates	Partner defects
Agent cooperates	3	−1
Agent defects	5	1

and it selects either a binary action or a graded investment depending on the regime.

In the binary game, both players choose cooperate or defect. Mutual cooperation gives the focal agent payoff 3, unilateral cooperation against a defecting partner gives −1, unilateral defection against a cooperating partner gives 5, and mutual defection gives 1. Graded regimes replace the binary cooperate/defect choice with a discretized investment level, but preserve the same social problem: the agent must decide how much to rely on a partner whose type and stance are hidden. Partner choice regimes add one more decision: before acting, the focal agent chooses which partner to engage.

For each partner k, the focal agent maintains a separate discrete active-inference model over partner type, partner stance, and own-action bookkeeping:

$$s_{t,k} = \left(s_{t,k}^{\text{type}}, s_{t,k}^{\text{stance}}, s_{t}^{\text{own}}\right). \tag{1}$$

The partner model is a discrete generative model

$$\mathcal{M}_k = \{A_k^{\text{action}}, A_k^{\text{payoff}}, B_k, C_k, D_k, E_k\}, \tag{2}$$

where A_k^{action} maps hidden type and stance to the partner's response, A_k^{payoff} maps own action and partner response to the payoff observation, B_k encodes state transitions, D_k stores prior beliefs, C_k encodes preferred observations, and E_k stores policy priors. Payoff is therefore represented as an observation modality inside the generative model rather than as an external reward cache. State and policy inference use the standard discrete active-inference machinery implemented in `pymdp` [4, 2, 6]. The trust-task code constructs the matrices, manages the partner bank, applies the external beta tracker, and logs readouts; it is not a custom replacement for the active-inference engine.

Partner choice is handled by evaluating partner-local policies and selecting among the resulting candidate interactions. In assigned-partner regimes, the agent chooses only the trust-game action. In partner-choice regimes, it first chooses whom to engage and then chooses an action toward that partner. This separates local social inference from the higher-level question of approach, avoidance, or probing: a partner can be predictable enough to avoid confidently, uncertain enough to probe, or reliable enough to return to.

The affective component is external to the POMDP state space. After interacting with partner k, the agent computes partner-action surprise

$$\epsilon_k = 1 - P(o_k^{\text{action}} = o_k^{\text{obs}} \mid q(s_{\text{type}}, s_{\text{stance}})). \tag{3}$$

Partner-local model $q_k(s_{\text{type}}, s_{\text{stance}})$	$\rightarrow$	Surprise update $\epsilon_k \rightarrow q(\beta_k)$	
	$\searrow$	$\downarrow$	
	Policy deployment $\gamma_k = \gamma_{\text{base}}/\mathbb{E}[\beta_k]$	$\rightarrow$	action toward partner k

Fig. 1: Partner-specific affective precision. Each partner has a local generative model; prediction error updates a partner-local beta estimate, which modulates the precision of future policies involving that partner.

This is a bounded prediction-error proxy in $[0, 1]$, not logarithmic surprisal. It is computed from partner action rather than payoff because partner action is the most direct readout of type-by-stance predictive fit.

The tracker converts this into affective charge

$$\phi_k = \alpha(\sigma_0^2 - \epsilon_k^2), \tag{4}$$

following the interpretation of affect as expected action precision and model fitness [7]. Positive charge means the partner model predicted better than the baseline variance σ_0^2; negative charge means it predicted worse. A categorical beta posterior $q(\beta_k)$ is updated over discrete beta levels $\{0.5, 0.67, 1.0, 1.5, 2.0\}$. The expected beta then modulates policy precision for future choices involving that partner:

$$\gamma_k = \gamma_{\text{base}}/\mathbb{E}_{q(\beta_k)}[\beta_k]. \tag{5}$$

Lower beta therefore sharpens the policy posterior; higher beta softens it. A tracked-only lesion keeps the beta update but fixes $\gamma_k = \gamma_{\text{base}}$, isolating deployment from belief updating.

The experiments are organized around five readouts rather than one payoff score. We first ask when the policy posterior is open enough for precision to matter. We then test whether precision follows predictive reliability more than reward, whether beta-to-gamma coupling changes deployment without improving beliefs, whether partner-local precision changes partner selection, and whether abrupt betrayal turns the same mechanism into misdeployment. Perturbation variants are treated as supplemental checks on beta dynamics rather than clinical validation.

3 Results

3.1 Model fitness is separable from reward

The reliability-versus-reward confirmation is the central mechanism result. The reward-cache interpretation predicts that precision should track payoff at least as strongly as predictive reliability. The model-fitness interpretation predicts the opposite: a predictable partner can support high precision even when interacting with that partner is not rewarding.

Across 30 seeds per variant and 200 rounds per seed, affective precision did not improve payoff: total payoff was 534.6 for the precision-modulated agent and 542.1 for the no-affect baseline, with a mean difference of -7.53 and bootstrap CI $[-27.88, 13.82]$. The absence of a payoff advantage is useful because it makes the reliability analysis a cleaner test of what the signal tracks.

The beta-derived precision signal tracked predictive reliability more strongly than reward. Across partner-seed units, $|\mathrm{corr}(\mathrm{precision}, \mathrm{surprise})| = 0.701$, while $|\mathrm{corr}(\mathrm{precision}, \mathrm{payoff})| = 0.419$. The corresponding seed-level reliability-over-reward summary was $+0.096$ with bootstrap CI $[0.027, 0.164]$. This dissociation supports the model-fitness interpretation: the affective signal estimates confidence in the partner model, not partner value.

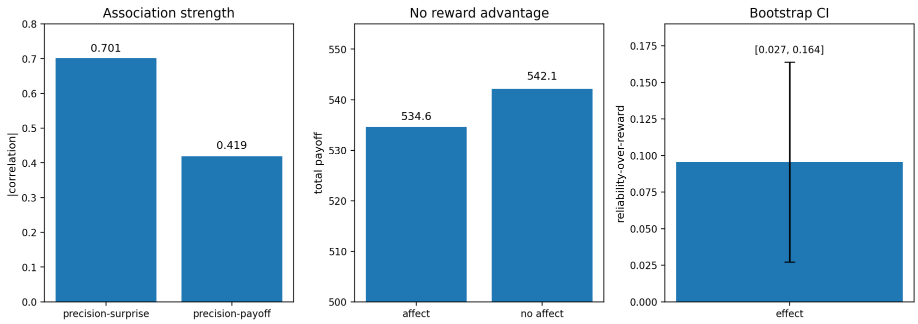

Fig. 2: Model-fitness readout. In the 30-seed reliability-versus-reward confirmation, partner-local precision tracks predictive surprise more strongly than realized payoff, while precision modulation has no total-payoff advantage.

3.2 Precision gates deployment in open policy regimes

Partner-local precision can only change behavior when several policies remain live enough for precision to matter. In a saturated regime, the best policy may already dominate the posterior; changing softmax precision then changes little. In the open graded regime, using five seeds per variant and 200 rounds per seed, partner-local precision changed behavior in the predicted direction. The precision-modulated agent reached total payoff 1884.6 with mean policy entropy 7.94, while the no-affect or tracked-only baseline reached 1859.4 with entropy 8.80. The difference was therefore $+25.2$ payoff and -0.86 entropy for precision modulation.

The tracked-only lesion clarifies the deployment claim. In the same open regime, the precision-modulated agent had joint belief accuracy 0.336 and stance accuracy 0.816, while the tracked-only lesion had joint accuracy 0.339 and stance accuracy 0.855. Belief readouts were therefore similar or slightly favored the lesion, but policy entropy and payoff favored the precision pathway. This is the intended deployment dissociation: the signal changes how beliefs are acted upon rather than simply improving belief accuracy.

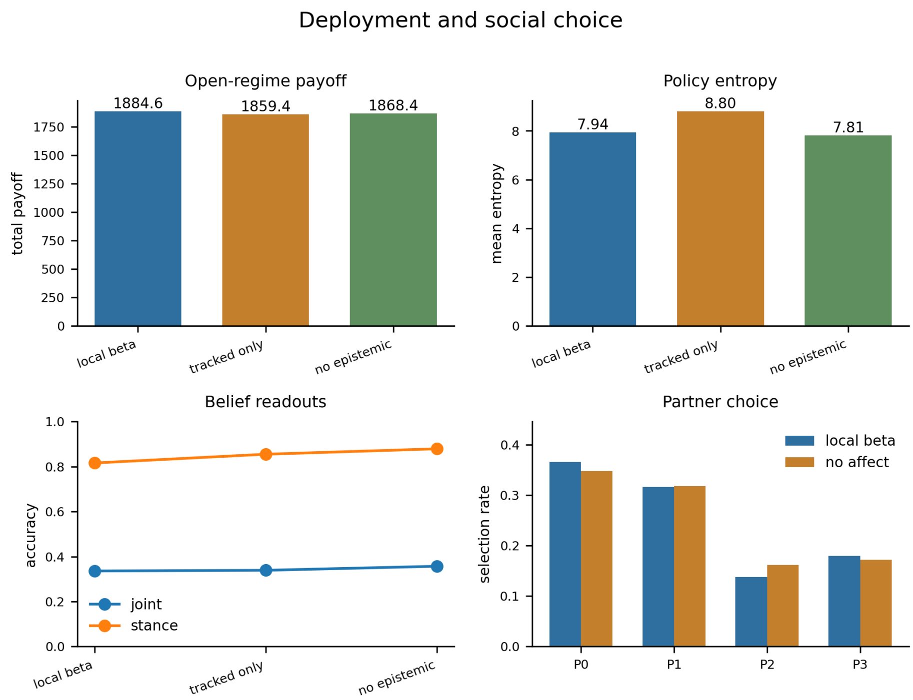

Fig. 3: Deployment and social choice readouts. In the open graded regime, local beta coupling changes payoff and policy entropy even when belief readouts remain similar. In the partner-choice regime, selection rates move before total payoff separates.

3.3 Partner choice changes before payoff

The partner-choice regime asks whether partner-local precision shapes approach, avoidance, and probing. Payoff is a blunt readout here because partner selection can change before total payoff separates. Across five seeds and 200 rounds per seed, payoff was nearly flat: 393.6 for precision modulation versus 393.2 for no affect. The policy distribution nevertheless changed. Mean policy entropy was 3.99 with precision modulation and 4.83 without it. Selection shifted toward the first partner (0.348 to 0.366) and away from the third partner (0.162 to 0.138). This supports the social-choice prediction in a bounded way: partner-local precision can shape approach and avoidance before total payoff becomes an informative readout.

3.4 Abrupt betrayal exposes a boundary condition

The abrupt-betrayal confirmation shows that the same precision channel can be harmful under social volatility. A partner's stance changes suddenly, creating a mismatch between the agent's accumulated model and the partner's new behavior. Across 30 seeds per variant and 120 rounds per seed, the precision-modulated agent obtained lower whole-run payoff than no-affect or tracked-only baselines: 1136.1 versus 1172.1, with mean difference -36.08, bootstrap CI $[-63.65, -10.93]$, and pairwise $p = 0.0169$. Policy entropy was lower under precision modulation (8.38 versus 8.74), showing that the channel was behaviorally active.

Post-switch readouts explain why this is not affective recovery. Precision modulation reduced mean reencounters with the switched partner (4.4 versus 6.1) and lowered reencounter selection rate (0.049 versus 0.067), but it did not improve the safety of returning: payoff on reencounter was 8.76 versus 8.91, and wrong-type rate on reencounter was 0.237 versus 0.165. The agent returned less often, but when it returned, it was not safer or better calibrated. Abrupt betrayal therefore shows precision-driven misdeployment: decisions become sharper before the partner model has recalibrated enough.

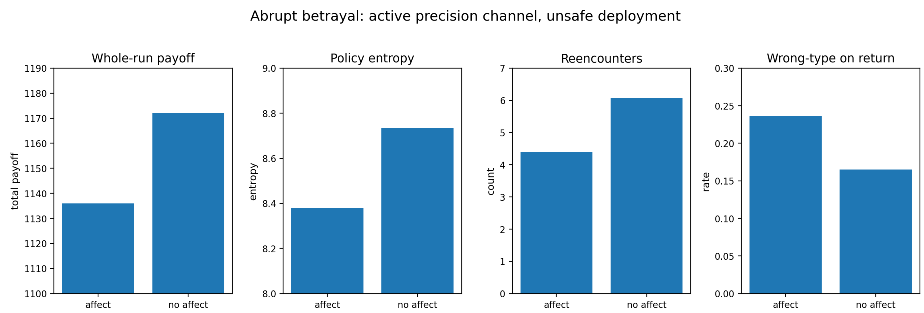

Fig. 4: Betrayal boundary condition. Abrupt betrayal makes partner-local precision behaviorally visible, but not safer: precision modulation lowers entropy and changes reallocation while reducing whole-run payoff.

3.5 Shock shape matters more than generic caution

The precision-sensitivity follow-up asks whether betrayal failure is simply excessive precision gain. In the abrupt sensitivity run, the no-affect/tracked-only baseline had payoff 1153.6 and entropy 8.80. Default precision modulation reached payoff 1140.4 and entropy 8.35 ($p = 0.370$ against no affect). A combined-caution variant raised entropy to 9.27 but lowered payoff to 1115.0 ($p = 0.003$ against no affect).

The gradual betrayal run changed the outcome. The no-affect/tracked-only baseline had payoff 1148.9 and entropy 8.76, while default precision modulation nearly matched it with payoff 1147.6, entropy 8.38, and $p = 0.906$. Combined caution again reduced payoff (1118.3, $p = 0.005$). Generic caution therefore does not rescue abrupt betrayal. The more informative boundary is shock shape: default partner-local precision is most harmful when a partner changes abruptly.

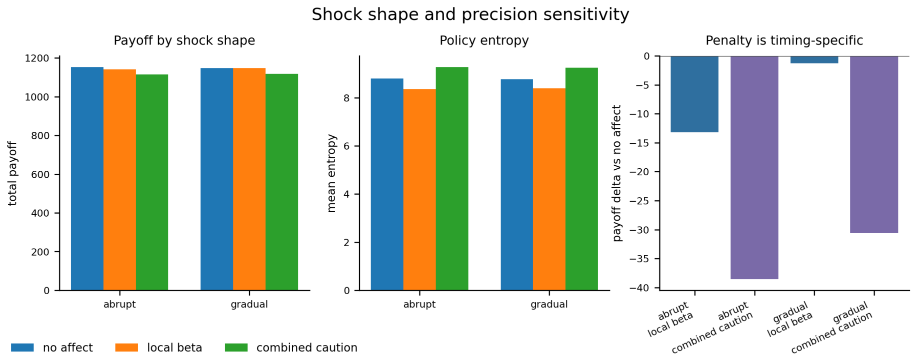

Fig. 5: Shock-shape sensitivity. Default local beta has a larger payoff penalty under abrupt betrayal than gradual betrayal, while a generic caution variant raises entropy but remains costly in both regimes.

3.6 Perturbations separate precision dynamics before interpretation

The perturbation runs are not clinical validation; they are a computational phenotyping check on the beta pathway. In open graded interaction, the low-gain variant compressed beta movement (mean beta range 0.149 versus 0.950 for default local beta), while high-gain and cautious-prior variants expanded it (1.318 and 1.435). In betrayal, the same qualitative ordering remained visible: low gain stayed blunted (0.132), while high gain and cautious prior remained more volatile (1.114 and 1.091). These differences are useful because they appear directly in the precision dynamics rather than only in payoff.

Behavioral readouts were less one-dimensional. Action churn stayed high across variants, and payoff did not rank variants by a simple “more beta movement is worse” rule. That pattern is the point of treating these as phenotypes rather than diagnoses: the perturbations define different beta dynamics, but the behavioral consequence depends on the regime in which those dynamics are deployed.

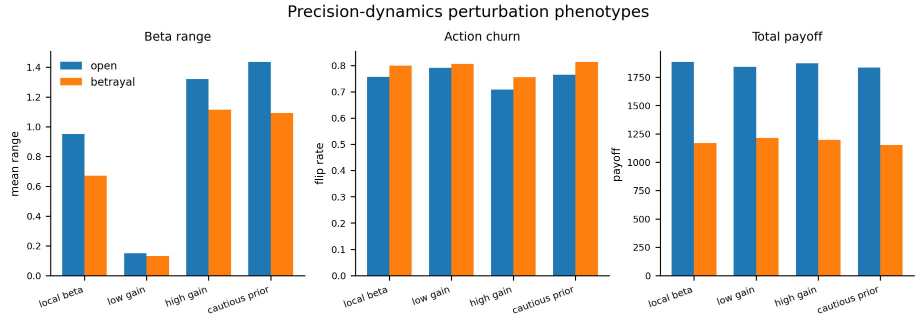

Fig. 6: Precision-dynamics perturbation phenotypes. Low-gain, high-gain, and cautious-prior variants separate clearly in beta movement, while action churn and payoff depend on the task regime. These variants are computational perturbations, not clinical models.

4 Discussion

These simulations support a constrained claim: affective precision can be factorized over social partners and used as a local model-fitness signal for social policy deployment. The signal is not equivalent to reward: in the reliability-versus-reward analysis, it tracks predictive surprise more strongly than payoff while producing no payoff advantage. It is also not merely belief accuracy: in the tracked-only lesion, belief readouts remain similar while policy entropy and payoff differ. Finally, it is not uniformly beneficial. The betrayal results show that abrupt volatility can make the same precision channel over-sharpen outdated assumptions.

This places the model between two simpler accounts. A reward-cache account cannot easily explain why precision follows prediction reliability more than reward, or why a predictable exploiter may still support confident action. A belief-only account cannot explain why similar partner-belief readouts can produce different policy entropy and payoff. Partner-local affective precision instead acts as a metacognitive deployment variable: it summarizes whether the agent's model of a particular partner is currently reliable enough to act on.

The work complements recent multi-agent active-inference models that factorize beliefs or reason about other agents [11, 10]. Our focus is different. We do not introduce a general theory-of-mind planning algorithm, nor do we assume that all agents share a generative model. We isolate a local precision channel that regulates how a focal agent deploys its own partner-specific social models. This is also distinct from active-inference accounts of trust in human-machine interaction [12]: here, the precision signal is not a literal trust scalar but confidence in the agent's model of a partner.

Betrayal as a useful failure. The abrupt-betrayal result is theoretically important because the precision channel moved behavior in the wrong direction for payoff. A weak or irrelevant channel would leave entropy, reencounter behavior, and wrong-type rates flat. Instead, precision modulation lowered entropy and changed reallocation while making returns less well calibrated. This suggests a timing problem: beta can alter policy precision before the slower partner-state posterior has fully reorganized around the new regime. In human terms, the agent becomes decisive too early; in active-inference terms, policy precision is misaligned with current posterior reliability.

Relation to trust. The beta signal should not be read as a literal trust scalar. Trust can mean willingness to rely on a partner, belief in benevolence, willingness to learn from testimony, or confidence in an interaction. This manuscript isolates one component: confidence in the focal agent's own predictive model of a partner. A reliable adversary can be modeled precisely and avoided; a valuable but erratic collaborator can remain worth engaging while being a poor basis for confident policy selection.

Limitations. The evidence is simulation-only and should not be read as validation of human affect or clinical phenotypes. The partners are scripted, which isolates the focal agent's mechanism but does not establish fully reciprocal multi-agent dynamics. The beta tracker is an external auxiliary filter rather than a higher-level hidden state inside the same variational model. The task also omits richer social structure such as communication, reputation, coalition formation, and recursive beliefs about what partners think the focal agent believes. Finally, the architecture is only beginning to be compared against a global-beta ablation. That comparison is the most direct next test of whether partner-specific affect is necessary, rather than simply useful.

Future work. The most important follow-up is a global-beta control that shares one affective precision estimate across partners while preserving the same beta-to-policy-precision mapping. A second follow-up is a focused shock-shape gradient that varies betrayal abruptness while holding the default precision channel fixed. More broadly, partner-local affective precision should be tested in richer reciprocal environments and, eventually, against human behavior.

5 Conclusion

Partner-specific affective precision provides a compact mechanism for linking social model fitness to action. In repeated trust-game simulations, it tracks predictive reliability more than partner reward, changes deployment in open policy regimes, and alters partner choice before payoff clearly moves. Under abrupt betrayal, however, the same mechanism can sharpen action around outdated assumptions and reduce payoff. Extending affective active inference to social settings therefore requires localizing precision over partner models and treating affect as a deployment signal that can both support and misdirect social action.

References

1. Berg, J., Dickhaut, J., McCabe, K.: Trust, reciprocity, and social history. Games and Economic Behavior **10**(1), 122–142 (1995). https://doi.org/10.1006/game.1995.1027
2. Da Costa, L., Parr, T., Sajid, N., Veselic, S., Neacsu, V., Friston, K.J.: Active inference on discrete state-spaces: A synthesis. Journal of Mathematical Psychology **99**, 102447 (2020). https://doi.org/10.1016/j.jmp.2020.102447
3. Friston, K.: The free-energy principle: a unified brain theory? Nature Reviews Neuroscience **11**, 127–138 (2010). https://doi.org/10.1038/nrn2787
4. Friston, K., FitzGerald, T., Rigoli, F., Schwartenbeck, P., Pezzulo, G.: Active inference: A process theory. Neural Computation **29**(1), 1–49 (2017). https://doi.org/10.1162/NECO_a_00912
5. Friston, K., Rigoli, F., Ognibene, D., Mathys, C., Fitzgerald, T., Pezzulo, G.: Active inference and epistemic value. Cognitive Neuroscience **6**(4), 187–214 (2015). https://doi.org/10.1080/17588928.2015.1020053
6. Heins, C., Millidge, B., Demekas, D., Klein, B., Friston, K., Couzin, I.D., Tschantz, A.: pymdp: A python library for active inference in discrete state spaces. Journal of Open Source Software **7**(73), 4098 (2022). https://doi.org/10.21105/joss.04098
7. Hesp, C., Smith, R., Parr, T., Allen, M., Friston, K.J., Ramstead, M.J.D.: Deeply felt affect: The emergence of valence in deep active inference. Neural Computation **33**(2), 398–446 (2021). https://doi.org/10.1162/neco_a_01341
8. King-Casas, B., Tomlin, D., Anen, C., Camerer, C.F., Quartz, S.R., Montague, P.R.: Getting to know you: reputation and trust in a two-person economic exchange. Science **308**(5718), 78–83 (2005). https://doi.org/10.1126/science.1108062
9. Parr, T., Pezzulo, G., Friston, K.J.: Active Inference: The Free Energy Principle in Mind, Brain, and Behavior. MIT Press (2022)
10. Pitliya, R.J., Çatal, O., Van de Maele, T., Pezzato, C., Verbelen, T.: Theory of mind using active inference: A framework for multi-agent cooperation (2025). https://doi.org/10.48550/arXiv.2508.00401
11. Ruiz-Serra, J., Sweeney, P., Harré, M.S.: Factorised active inference for strategic multi-agent interactions. In: Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems. pp. 1793–1802. International Foundation for Autonomous Agents and Multiagent Systems (2025). https://doi.org/10.5555/3709347.3743815
12. Schoeller, F., Miller, M., Salomon, R., Friston, K.J.: Trust as extended control: Human-machine interactions as active inference. Frontiers in Systems Neuroscience **15**, 669810 (2021). https://doi.org/10.3389/fnsys.2021.669810